Digital Evaluation Copy

For Teachers & Departments

Further Mathematics for CCEA A-Level: Discrete and Decision Mathematics

Complete AS and A2 Course

CCEA GCE Further Mathematics • ISBN 978-1-0666184-9-1

by Isaac Teng — MSci Mathematics & Physics, UCL; Founder, IREducation Inc.

What this evaluation copy contains

These are complete, unaltered excerpts from the full textbook. Published page and chapter numbers are retained so that the sample can be compared directly with the complete book.

- **Front matter** — Introduction, About the Author, and the full Contents.
- **Chapter 3** — Recurrences and Boolean Algebra (pp. 63–84), with solutions to Exercises 3A–3C. End-of-chapter mixed-exercise solutions are reserved for the complete book.
- **Appendix A** — the calculator guide, Sections A.1–A.4: counting and restricted assignments, recurrence relations, PERT calculations, and simplex and cycle-index arithmetic.
- **AS1 Practice Set** — the first AS Discrete and Decision Mathematics practice set, with its matching solutions.

The full book at a glance

- 240 pages and six chapters covering the complete CCEA Discrete and Decision Mathematics option at AS and A2.
- Six practice sets — three at AS and three at A2 — with complete worked solutions.
- A calculator guide, a discrete and decision reference and per-chapter worked answers.

© 2026 IREducation Inc. — Not for redistribution.

Further Mathematics for CCEA A-Level: Discrete and Decision Mathematics

Complete AS and A2 Course

Isaac Teng Chit Hang

Further Mathematics for CCEA A-Level: Discrete and Decision Mathematics
Complete AS and A2 Course

Copyright © 2026 IREducation Inc. All rights reserved.

First edition, 2026. Written and published by Isaac Teng Chit Hang.

This is an independent publication. It is not endorsed by, approved by, or affiliated with the Council for the Curriculum, Examinations and Assessment (CCEA).

A2 Discrete and Decision Mathematics assumes the AS section as well as the compulsory pure units.

ISBN-13: 978-1-0666184-9-1

Introduction

This textbook is the **Discrete and Decision Mathematics** volume in the IRMaths series for **CCEA GCE Further Mathematics**. Companion volumes cover compulsory Pure Mathematics and the other applied options. It covers this option at AS and A2 and is intended for use alongside the relevant Pure Mathematics material.

The six chapters follow the CCEA pathway. Chapters 1–3 cover AS groups, graph algorithms, recurrences and Boolean algebra. Chapters 4–6 develop A2 counting, advanced graph theory, optimisation, generating functions and symmetry enumeration. They assume familiarity with the structures, algorithms and notation developed at AS.

The aim is to guide students from basic discrete reasoning to confident work with unfamiliar problems. A complete solution must select an appropriate model, recognise its assumptions, apply the mathematics accurately and interpret the result in context. The text emphasises precise notation, justified method selection and clear conclusions as well as calculation.

Each chapter begins with its learning goals and develops the ideas through explanation, key results and worked examples. Section exercises provide focused practice; the end-of-chapter mixed exercises bring methods together and progress from routine fluency to multi-step modelling, interpretation and the diagnosis of common errors. Fully worked answers allow students to check both their result and the structure of their reasoning.

The six practice sets—three at AS and three at A2—provide consolidation once the relevant content is secure. They should be used alongside the latest official specification, specimen materials, past papers and instructions to candidates.

The calculator guide gives model-specific key sequences for the Casio fx-991EX, fx-991CW and fx-CG50. Recurrence relations, algorithm steps, substitutions and conclusions should still be shown. The Discrete and Decision Reference appendix gathers the principal notation and results.

The book can be used in class, for independent study or as a revision reference, and alongside the IRMaths platform (`irmaths.ireducation.org`), which provides further resources across the CCEA Further Mathematics series.

Specifications, assessment arrangements and calculator software can change, so students and teachers should consult current CCEA documents. Corrections and suggestions are welcome at `isaacteng@ireducation.org`; please identify the edition, page and exercise number.

Thank you for choosing this book. I hope that it helps you develop not only the techniques required for the examination, but also the judgement and clarity that make discrete mathematics useful beyond it.

A Note on Beyond the Syllabus Content

Purple Beyond the Syllabus boxes appear only where they illuminate an assessed idea or prepare for later mathematics. They are optional and are not required for the CCEA examination.

About the Author

Isaac Teng is the founder of IREducation Inc., an ed-tech organization based in Hong Kong SAR. He is also an MSci Mathematics and Physics student at University College London (UCL), and the lead developer for the IREducation platform, including IRMaths.

Contents

Introduction **3**

1 Groups **9**
1.1 The Group Axioms . . . 10
1.2 Group Order and Element Period . . . 17
1.3 Subgroups and Lagrange's Theorem . . . 19
1.4 Cyclic Groups and Generators . . . 25
1.5 Isomorphic Groups . . . 28

2 Graph Theory and Algorithms **35**
2.1 Core Graph Concepts and Special Graphs . . . 36
2.2 Traversability . . . 41
2.3 Weighted Graphs, Digraphs, Trees and Prim's Algorithm . . . 44
2.4 Shortest Paths . . . 49
2.5 Tree Traversals . . . 52
2.6 Critical-Path Analysis . . . 54

3 Recurrences and Boolean Algebra **63**
3.1 Modelling with Recurrence Relations . . . 63
3.2 Solving Homogeneous Constant-Coefficient Linear Recurrences . . . 69
3.3 Boolean Algebra and Propositional Logic . . . 75

4 Advanced Counting and Graph Theory **85**
4.1 Inclusion-Exclusion and Derangements . . . 86
4.2 Restricted Positions and Rook Polynomials . . . 90
4.3 Graph Colouring (Vertex and Edge) . . . 95
4.4 Network Flows: Cutsets and the Max-Flow Min-Cut Theorem . . . 98
4.5 Bipartite Graphs, Matchings, and Hall's Marriage Theorem . . . 102

5 Advanced Algorithms and Generating Functions **109**
5.1 The Nearest Neighbour Algorithm . . . 110
5.2 PERT Analysis . . . 114
5.3 Simplex Tableaux for Two-Variable Linear Programming . . . 119
5.4 Generating Functions . . . 126

6 Symmetry Groups and Pólya Enumeration **137**
6.1 Symmetry Groups . . . 138
6.2 The Cycle Index . . . 143
6.3 Pólya's Enumeration Theorem . . . 148
6.4 Pattern Inventories . . . 151

Appendices **159**

A Calculator Guide **161**
A.1 Counting and Restricted Assignments . . . 161
A.2 Recurrence Relations . . . 162
A.3 PERT Means, Variances and Probabilities . . . 163
A.4 Simplex and Cycle-Index Arithmetic . . . 163

B Discrete and Decision Reference **165**
B.1 Groups . . . 165
B.2 Graphs and AS Algorithms . . . 166
B.3 Recurrences and Boolean Algebra . . . 167
B.4 Counting, Colouring, Flows and Matchings . . . 167
B.5 A2 Algorithms and Generating Functions . . . 168
B.6 Symmetry and Enumeration . . . 169
B.7 Statistical Tables . . . 170

C External Resources **171**

Practice Sets **173**

AS Discrete and Decision Mathematics Practice Set 1 **175**

AS Discrete and Decision Mathematics Practice Set 2 **177**

AS Discrete and Decision Mathematics Practice Set 3 **179**

A2 Discrete and Decision Mathematics Practice Set 1 **181**

A2 Discrete and Decision Mathematics Practice Set 2 **183**

A2 Discrete and Decision Mathematics Practice Set 3 **185**

Answers **187**

Chapter 3

Recurrences and Boolean Algebra

After completing this chapter you should be able to:

- Model discrete processes using recurrence relations and initial conditions.
- Solve first-order and second-order homogeneous constant-coefficient linear recurrence relations.
- Derive closed-form expressions for Fibonacci-type sequences involving exact surds.
- Construct systematic truth tables for compound propositions involving at most three variables.
- Use truth tables to prove or disprove the logical equivalence of propositions.

Chapter route

Many systems in science, finance and computing change in distinct steps rather than continuously. The first part of this chapter shows how to translate step-by-step rules into recurrence relations paired with initial conditions. The second part develops an algebraic method for turning linear homogeneous recurrence relations into explicit closed-form formulae for the nth term, including sequences of Fibonacci type whose roots are irrational. The final part introduces propositional logic and Boolean operators, showing how truth tables provide an exhaustive proof of logical equivalence for statements with up to three variables.

Prerequisites

Before starting this chapter, you should be confident with:

- Sequences and series, including arithmetic and geometric progressions (AS Mathematics).
- Solving quadratic equations, including expressing exact roots in surd form (AS Mathematics).
- Solving pairs of simultaneous linear equations in two unknowns (AS Mathematics).

3.1 Modelling with Recurrence Relations

3.1.1 Sequences, Closed Forms and Recursive Definitions

A **sequence** is an ordered list of numbers, such as $2, 5, 8, 11, \ldots$ or $1, 1, 2, 3, 5, 8, \ldots$. Each number in the list is called a **term**, and we use subscript notation such as $u_0, u_1, u_2, \ldots, u_n$ to indicate the position of

each term.

There are two standard ways to specify a sequence:

- A **closed-form formula** (or position-to-term rule) expresses u_n directly as a function of the index n, such as $u_n = 3n - 1$. This allows any term, such as u_{100}, to be calculated in a single substitution.
- A **recurrence relation** (or term-to-term rule) expresses each new term as a function of one or more preceding terms, such as $u_n = u_{n-1} + 3$ or $u_{n+1} = u_n + 3$.

When modelling a discrete process, we usually observe how the system changes from one step to the next—for instance, how a bank balance changes over a single month or how a biological population changes from one breeding season to the next. Because the transition rule between consecutive steps is directly known, it is usually much easier to formulate a recurrence relation first than to write down a closed-form formula immediately.

A recurrence relation on its own does not specify a unique sequence. For example, the sequences

$$5, 8, 11, 14, \ldots \quad \text{and} \quad 10, 13, 16, 19, \ldots$$

both satisfy $u_n = u_{n-1} + 3$. To identify one specific sequence from the infinite family of sequences obeying the recurrence rule, we must supply **initial conditions**: the numerical values of the starting term(s).

Key Point

Order of a Recurrence Relation and Initial Conditions:

- A **first-order** recurrence relation expresses u_n in terms of only the immediately preceding term u_{n-1}:
$$u_n = f(u_{n-1})$$
A first-order recurrence relation requires **one** initial condition (such as u_0 or u_1) to define a unique sequence.
- A **second-order** recurrence relation expresses u_n in terms of the two immediately preceding terms u_{n-1} and u_{n-2}:
$$u_n = f(u_{n-1}, u_{n-2})$$
A second-order recurrence relation requires **two** initial conditions (such as u_0 and u_1, or u_1 and u_2) to define a unique sequence.

Common Misconception

Be careful with the starting index in applied models. When n measures the number of completed time periods (such as years or months elapsed), the initial state at the start of the process corresponds to $n = 0$ (u_0), so u_1 represents the state at the *end* of the first period. When n counts discrete events or positions (such as the nth tablet taken or the nth term in a list), the sequence normally starts at $n = 1$ (u_1). Always check what the subscript n represents before substituting.

3.1.2 Modelling Population Dynamics and Resource Management

Many biological populations reproduce in annual cycles rather than continuously. If a population grows by a fixed percentage each season and a fixed number of individuals are added or removed, the population after n seasons can be modelled by a first-order linear recurrence relation of the form

$$P_n = rP_{n-1} + d$$

where r is the growth multiplier and d is the net number added (if $d > 0$) or removed (if $d < 0$) each period.

Example 1

Question: A conservation team is modelling the population of a rare species of bird on an isolated island. Initially, there are 50 birds on the island. Each year, the population increases by 15% through breeding, and at the end of the year 4 birds are removed from the island to establish a mainland breeding programme.

(a) Write down a recurrence relation for P_n, the population of birds on the island after n years, and state the initial condition.

(b) Use the recurrence relation to estimate the bird population after 3 years.

(c) If the conservation team instead removed k birds at the end of each year, find the smallest integer value of k for which the island population would decrease from year to year.

Solution:

(a) A 15% annual increase corresponds to a multiplier of $1 + 0.15 = 1.15$. At the end of year n, the previous year's population P_{n-1} is multiplied by 1.15, and then 4 birds are subtracted:

$$P_n = 1.15P_{n-1} - 4 \quad (n \geq 1)$$

Since there are 50 birds initially (when $n = 0$), the initial condition is $P_0 = 50$.

(b) Calculating the terms step by step from $P_0 = 50$:

$$P_1 = 1.15(50) - 4 = 57.5 - 4 = 53.5$$
$$P_2 = 1.15(53.5) - 4 = 61.525 - 4 = 57.525$$
$$P_3 = 1.15(57.525) - 4 = 66.15375 - 4 = 62.15375$$

Rounding to the nearest whole number for a count of living birds, the estimated population after 3 years is 62 birds.

(c) With k birds removed each year, $P_1 = 1.15(50) - k = 57.5 - k$. For the population to decrease from its initial value of 50, we require $P_1 < P_0$:

$$57.5 - k < 50 \implies k > 7.5$$

Since k must be an integer, the smallest number of birds that would cause the population to decrease is $k = 8$.

3.1.3 Modelling Financial Processes and Order of Operations

Recurrence relations are widely used in financial mathematics to model savings accounts, annuities and loans where compound interest is combined with regular deposits or repayments. When translating a worded description into a recurrence relation, pay close attention to the order in which events occur within each time period.

Suppose an account with balance V_{n-1} earns interest at rate i per period and receives a deposit of D:

- If interest is calculated on the balance V_{n-1} **before** the deposit D is added at the end of the period,

then

$$V_n = (1+i)V_{n-1} + D.$$

- If the deposit D is paid in at the **start** of the period so that both V_{n-1} and D earn interest during that period, then

$$V_n = (1+i)(V_{n-1} + D) = (1+i)V_{n-1} + (1+i)D.$$

Example 2

Question: A graduate opens a savings account with an initial deposit of £1000. At the end of each month, the account earns 0.5% compound interest on the balance held during that month, and the graduate immediately pays in an additional £200. Let V_n be the balance in pounds at the end of n months.

(a) Write down a recurrence relation for V_n and state the initial condition.

(b) Find the account balance at the end of 3 months, giving your answer to the nearest penny.

Solution:

(a) A monthly interest rate of 0.5% gives a multiplier of $1 + 0.005 = 1.005$. Because the £200 deposit is added at the end of each month after interest has been applied to V_{n-1}, the recurrence relation is

$$V_n = 1.005V_{n-1} + 200 \quad (n \geq 1)$$

with initial condition $V_0 = 1000$.

(b) Iterating from $V_0 = 1000$:

$$V_1 = 1.005(1000) + 200 = 1005 + 200 = 1205$$
$$V_2 = 1.005(1205) + 200 = 1211.025 + 200 = 1411.025$$
$$V_3 = 1.005(1411.025) + 200 = 1418.080125 + 200 = 1618.080125$$

Rounding to the nearest penny, the balance after 3 months is £1618.08.

3.1.4 Constructing Second-Order Recurrence Models

A second-order recurrence relation arises when the state at step n depends on two distinct ways of reaching that state—one from step $n-1$ and one from step $n-2$.

Example 3

Question: A garden path of length n metres ($n \geq 1$) and width 1 metre is to be paved in a single row using grey stone slabs of length 1 m and red stone slabs of length 2 m. Let u_n denote the number of different arrangements of slabs that form a path of length n metres.

(a) By listing the possible arrangements, find the values of u_1 and u_2.

(b) By considering the length of the first slab laid at the start of the path, explain why $u_n = u_{n-1} + u_{n-2}$ for $n \geq 3$.

(c) Hence calculate the number of different arrangements for a path of length 6 metres.

Solution:

(a) For $n = 1$ m, only a single 1 m grey slab can be used: [1]. Thus $u_1 = 1$.
For $n = 2$ m, we can use either two 1 m grey slabs $[1, 1]$ or one 2 m red slab [2]. Thus $u_2 = 2$.

(b) Consider a path of length $n \geq 3$ metres. The first slab laid must have length either 1 m or 2 m:

- If the first slab has length 1 m, the remaining length of path to be paved is $(n - 1)$ m, which can be completed in u_{n-1} ways.
- If the first slab has length 2 m, the remaining length of path to be paved is $(n - 2)$ m, which can be completed in u_{n-2} ways.

Because these two cases do not overlap and cover all possibilities, the total number of arrangements is

$$u_n = u_{n-1} + u_{n-2} \quad (n \geq 3).$$

(c) Using the initial conditions $u_1 = 1$ and $u_2 = 2$ with the recurrence relation:

$$\begin{aligned} u_3 &= u_2 + u_1 = 2 + 1 = 3 \\ u_4 &= u_3 + u_2 = 3 + 2 = 5 \\ u_5 &= u_4 + u_3 = 5 + 3 = 8 \\ u_6 &= u_5 + u_4 = 8 + 5 = 13 \end{aligned}$$

There are 13 different arrangements for a path of length 6 metres.

Beyond the Syllabus: The Tower of Hanoi and Algorithmic Complexity

In computer science, recurrence relations are used to count the number of operations required by recursive algorithms. A classic example is the **Tower of Hanoi** puzzle, introduced by Édouard Lucas in 1883. Three vertical pegs hold n discs of different diameters, stacked on one peg in decreasing order of size from bottom to top. The goal is to transfer the entire stack to another peg by moving one disc at a time, never placing a larger disc on top of a smaller one.

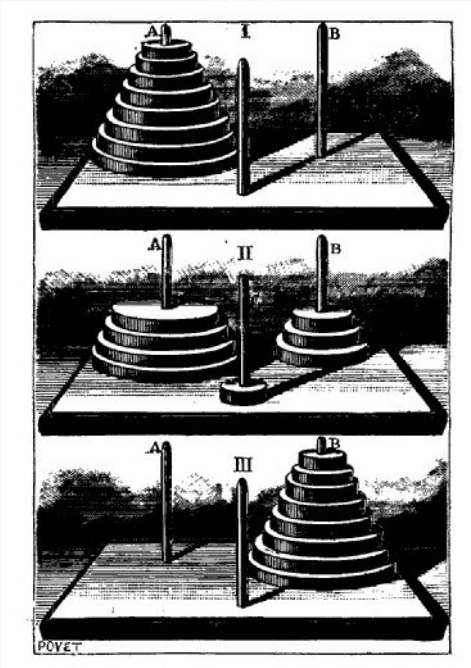


Figure 3.1: An 1883 illustration of the Tower of Hanoi. Lucas first sold the puzzle under the pseudonym “N. Claus de Siam”, an anagram of “Lucas d’Amiens”.

Let T_n be the minimum number of moves needed to transfer n discs. To move the largest (bottom) disc, the top $n-1$ smaller discs must first be transferred to the spare peg (requiring T_{n-1} moves). The largest disc is then moved to the target peg (1 move), and finally the $n-1$ smaller discs are transferred on top of it (requiring a further T_{n-1} moves). This structure yields the first-order recurrence relation

$$T_n = 2T_{n-1} + 1 \quad (n \geq 2), \quad T_1 = 1.$$

Generating the first few terms gives $T_1 = 1$, $T_2 = 3$, $T_3 = 7$, $T_4 = 15$, which leads to the closed-form solution $T_n = 2^n - 1$.

Exercise 3A

1 A sequence is defined by the recurrence relation $u_n = 3u_{n-1} - 2$ for $n \geq 2$.

(a) Given the initial condition $u_1 = 4$, find the values of u_2, u_3 and u_4. **[2 marks]**

(b) A second sequence satisfies the same recurrence relation $v_n = 3v_{n-1} - 2$ for $n \geq 2$, but has $v_3 = 19$. Find the value of the initial term v_1. **[2 marks]**

2 An algae bloom initially covers an area of 12 m^2 on the surface of a pond. Each week, the area covered by the algae increases by a constant factor $k > 0$, so that the area A_n m^2 at the end of week n satisfies $A_n = kA_{n-1}$. At the end of week 2, the area covered is 27 m^2.

(a) State the initial condition A_0 and find the value of the constant k. **[2 marks]**

(b) Calculate the area covered by the algae at the end of week 5. **[2 marks]**

3 A patient takes a tablet containing 50 mg of an active drug at the same time each morning. Over the 24 hours between doses, the kidneys remove 40% of the drug present in the bloodstream. Let D_n be the mass, in mg, of the drug in the bloodstream immediately after the nth tablet is taken.

(a) Given that $D_1 = 50$, write down a recurrence relation for D_n in terms of D_{n-1} for $n \geq 2$. **[2 marks]**

(b) Calculate the mass of the drug in the bloodstream immediately after the 4th tablet is taken. **[2 marks]**

(c) After many days, the mass of the drug immediately after a dose approaches a steady limit L mg satisfying $L = 0.6L + 50$. Find the value of L and hence state whether the mass of the drug in the bloodstream ever exceeds 130 mg. **[2 marks]**

4 ⋆ A managed forest initially contains 4000 mature trees. Each year, natural regeneration increases the number of trees by 5%, and a forestry company fells h trees for timber. Let T_n be the number of trees in the forest after n years.

(a) Model A assumes that the h trees are felled at the end of each year, after the 5% growth has occurred. Write down a recurrence relation for T_n under Model A and state the initial condition. **[2 marks]**

(b) Model B assumes that the h trees are felled at the start of each year, and the 5% growth then applies to the remaining trees. Show that Model B gives the recurrence relation $T_n = 1.05T_{n-1} - 1.05h$. **[2 marks]**

(c) Using Model A, find the number of trees felled per year, h, given that there are 3959 trees in the forest after 2 years. **[3 marks]**

5 A robotic inspection drone climbs a flight of n steps ($n \geq 1$). In a single move, the drone can ascend either 1 step or 2 steps. Let s_n be the total number of distinct sequences of moves by which the drone can ascend n steps.

(a) Write down the values of s_1 and s_2. **[1 mark]**

(b) By considering the drone's first move, explain why $s_n = s_{n-1} + s_{n-2}$ for $n \geq 3$, and hence calculate s_7. **[3 marks]**

3.2 Solving Homogeneous Constant-Coefficient Linear Recurrences

3.2.1 Structure and Classification of Linear Recurrences

Step-by-step iteration is practical for finding u_3 or u_4, but it becomes slow and prone to rounding error if we need u_{50} or want to understand the long-term behaviour of a sequence as $n \to \infty$. For an important class of recurrence relations, we can bypass iteration altogether by finding a closed-form algebraic formula for u_n in terms of n.

In this section we solve **homogeneous linear recurrence relations with constant coefficients**. Each word in this description places a specific condition on the equation:

- **Linear:** Each sequence term $(u_n, u_{n-1}, u_{n-2}, \dots)$ occurs to the first power only, with no products of terms such as $u_{n-1}u_{n-2}$ and no non-linear functions such as u_{n-1}^2 or $\sqrt{u_{n-1}}$.
- **Constant-coefficient:** The coefficients multiplying the sequence terms are fixed numbers rather than functions of n. For example, $u_n = 5u_{n-1} - 6u_{n-2}$ has constant coefficients, whereas $u_n = nu_{n-1}$ does not.
- **Homogeneous:** When all sequence terms are collected on the left-hand side, the right-hand side is zero. Every term in the equation contains a member of the sequence. For instance, $u_n - 3u_{n-1} = 0$ is homogeneous, whereas $u_n - 3u_{n-1} = -2$ is non-homogeneous because of the standalone constant -2.

3.2.2 From First-Order Recurrences to the Auxiliary Equation

To see why exponential trial solutions work, consider first a first-order homogeneous linear recurrence relation with constant coefficients:

$$au_n + bu_{n-1} = 0 \quad (a \neq 0,\ b \neq 0).$$

Rearranging gives $u_n = ru_{n-1}$, where $r = -\frac{b}{a}$ is a non-zero constant. This is a geometric progression with common ratio r: each step multiplies the previous term by r, so starting from $u_0 = A$ gives

$$u_1 = Ar, \quad u_2 = Ar^2, \quad u_3 = Ar^3, \quad \ldots, \quad u_n = Ar^n.$$

Notice that if we substitute the trial form $u_n = \lambda^n$ $(\lambda \neq 0)$ into $au_n + bu_{n-1} = 0$, we obtain

$$a\lambda^n + b\lambda^{n-1} = 0.$$

Dividing through by λ^{n-1} gives $a\lambda + b = 0$, so $\lambda = -\frac{b}{a} = r$, recovering the general solution $u_n = A\lambda^n$.

Key Point

Second-order homogeneous linear recurrence. The standard form, with constant coefficients, is

$$au_n + bu_{n-1} + cu_{n-2} = 0 \quad (a \neq 0,\ c \neq 0).$$

Apply the trial solution $u_n = \lambda^n$ $(\lambda \neq 0)$. Substituting $u_n = \lambda^n$, $u_{n-1} = \lambda^{n-1}$ and $u_{n-2} = \lambda^{n-2}$ yields

$$a\lambda^n + b\lambda^{n-1} + c\lambda^{n-2} = 0.$$

Since $\lambda \neq 0$, we can divide every term by the lowest power, λ^{n-2}, to obtain the quadratic equation

$$a\lambda^2 + b\lambda + c = 0.$$

This quadratic is called the **auxiliary equation** (or characteristic equation) of the recurrence relation.

Why can we combine two basic solutions to form a general solution containing two arbitrary constants? Suppose p_n and q_n are any two sequences that satisfy $au_n + bu_{n-1} + cu_{n-2} = 0$. For any constants A and B, let $u_n = Ap_n + Bq_n$. Substituting u_n into the left-hand side of the recurrence gives

$$\begin{aligned} &a(Ap_n + Bq_n) + b(Ap_{n-1} + Bq_{n-1}) + c(Ap_{n-2} + Bq_{n-2}) \\ &\quad = A(ap_n + bp_{n-1} + cp_{n-2}) + B(aq_n + bq_{n-1} + cq_{n-2}) \\ &\quad = A(0) + B(0) = 0. \end{aligned}$$

This property is called the **principle of superposition**: any linear combination of solutions to a linear homogeneous recurrence relation is also a solution.

Key Point

Solving Second-Order Homogeneous Linear Recurrence Relations:
To solve $au_n + bu_{n-1} + cu_{n-2} = 0$ (or equivalently $au_{n+2} + bu_{n+1} + cu_n = 0$):

1. Substitute $u_n = \lambda^n$ $(\lambda \neq 0)$ to form the **auxiliary equation**:

$$a\lambda^2 + b\lambda + c = 0.$$

2. Solve the auxiliary equation for its roots λ_1 and λ_2:

- **Case 1: Distinct real roots** ($\lambda_1 \neq \lambda_2$). The general solution is

$$u_n = A\lambda_1^n + B\lambda_2^n.$$

- **Case 2: Repeated real root** ($\lambda_1 = \lambda_2 = \lambda$). The general solution is

$$u_n = (A + Bn)\lambda^n.$$

3. If initial conditions are given, substitute them into the general solution to form and solve simultaneous linear equations for the constants A and B, giving the **particular solution**.

Common Misconception

Two errors frequently occur when setting up and solving the auxiliary equation:

1. **Sign errors before forming the quadratic:** If a recurrence is given as $u_n = 7u_{n-1} - 10u_{n-2}$, always rearrange all terms to one side first ($u_n - 7u_{n-1} + 10u_{n-2} = 0$) before writing $\lambda^2 - 7\lambda + 10 = 0$. Writing $\lambda^2 = 7\lambda - 10$ is also safe.

2. **Omitting the factor of n for a repeated root:** When $\lambda_1 = \lambda_2 = \lambda$, writing $u_n = A\lambda^n + B\lambda^n = (A+B)\lambda^n$ collapses the two constants into a single constant $C = A + B$, which cannot satisfy two independent initial conditions. You must include the second independent solution $n\lambda^n$, giving $u_n = (A + Bn)\lambda^n$.

3.2.3 Case 1: Distinct Real Roots

When the discriminant $b^2 - 4ac > 0$, the auxiliary equation has two distinct real roots λ_1 and λ_2, giving two independent geometric solutions λ_1^n and λ_2^n.

Example 4

Question: A sequence is defined by the recurrence relation $u_n = 7u_{n-1} - 10u_{n-2}$ for $n \geq 2$, subject to the initial conditions $u_0 = 4$ and $u_1 = 17$.

(a) Find the general solution for u_n.

(b) Find the particular solution satisfying the initial conditions, and verify your formula for $n = 2$.

Solution:

(a) Rearrange the recurrence relation into standard homogeneous form:

$$u_n - 7u_{n-1} + 10u_{n-2} = 0$$

Substituting $u_n = \lambda^n$ gives the auxiliary equation:

$$\lambda^2 - 7\lambda + 10 = 0$$

Factorising:

$$(\lambda - 2)(\lambda - 5) = 0 \implies \lambda_1 = 2, \quad \lambda_2 = 5$$

Since the roots are real and distinct, the general solution is

$$u_n = A(2)^n + B(5)^n.$$

(b) Substitute the initial conditions $u_0 = 4$ and $u_1 = 17$:

$$n = 0: \quad A(2)^0 + B(5)^0 = 4 \implies A + B = 4 \qquad \text{— (Equation 1)}$$
$$n = 1: \quad A(2)^1 + B(5)^1 = 17 \implies 2A + 5B = 17 \qquad \text{— (Equation 2)}$$

From Equation 1, $A = 4 - B$. Substituting into Equation 2 gives

$$2(4 - B) + 5B = 17 \implies 8 + 3B = 17 \implies 3B = 9 \implies B = 3.$$

Then $A = 4 - 3 = 1$. The particular solution is therefore

$$u_n = 2^n + 3(5)^n.$$

Check ($n = 2$): From the recurrence relation, $u_2 = 7u_1 - 10u_0 = 7(17) - 10(4) = 119 - 40 = 79$. From the closed-form formula, $u_2 = 2^2 + 3(5)^2 = 4 + 75 = 79$, confirming the solution.

3.2.4 Fibonacci-Type Sequences and Exact Surd Roots

Key Point

The **Fibonacci sequence** is defined by the second-order linear homogeneous recurrence relation

$$F_n = F_{n-1} + F_{n-2} \quad (n \geq 2)$$

with initial conditions $F_0 = 0$ and $F_1 = 1$, generating the terms $0, 1, 1, 2, 3, 5, 8, 13, 21, \ldots$.

More generally, any sequence satisfying a recurrence of the form $u_n = pu_{n-1} + qu_{n-2}$ whose auxiliary equation has irrational surd roots (or the same recurrence $u_n = u_{n-1} + u_{n-2}$ with different initial conditions, such as in Worked Example 3) is solved by the same method, keeping all roots and constants in exact surd form.

Example 5

Question: A sequence is defined by $u_n = u_{n-1} + u_{n-2}$ for $n \geq 2$, with initial conditions $u_0 = 0$ and $u_1 = 1$. Find a closed-form algebraic expression for u_n in terms of n.
Solution: Rearranging the recurrence relation gives

$$u_n - u_{n-1} - u_{n-2} = 0.$$

The auxiliary equation is

$$\lambda^2 - \lambda - 1 = 0.$$

Using the quadratic formula with $a = 1$, $b = -1$ and $c = -1$:

$$\lambda = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)} = \frac{1 \pm \sqrt{5}}{2}.$$

Since the two roots $\lambda_1 = \frac{1+\sqrt{5}}{2}$ and $\lambda_2 = \frac{1-\sqrt{5}}{2}$ are real and distinct, the general solution is

$$u_n = A\left(\frac{1 + \sqrt{5}}{2}\right)^n + B\left(\frac{1 - \sqrt{5}}{2}\right)^n.$$

Applying the initial condition $u_0 = 0$:

$$A(1) + B(1) = 0 \implies B = -A.$$

Applying the initial condition $u_1 = 1$ and substituting $B = -A$:

$$A\left(\frac{1+\sqrt{5}}{2}\right) - A\left(\frac{1-\sqrt{5}}{2}\right) = 1.$$

Factorising A and simplifying the bracket:

$$A\left(\frac{1+\sqrt{5}-1+\sqrt{5}}{2}\right) = 1 \implies A\left(\frac{2\sqrt{5}}{2}\right) = 1 \implies A\sqrt{5} = 1 \implies A = \frac{1}{\sqrt{5}}.$$

Hence $B = -\frac{1}{\sqrt{5}}$, and the closed-form solution (known as **Binet's formula**) is

$$u_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n.$$

3.2.5 Case 2: Repeated Real Roots

When the discriminant $b^2 - 4ac = 0$, the auxiliary equation $a\lambda^2 + b\lambda + c = 0$ factorises as $a(\lambda - \lambda_0)^2 = 0$, yielding a single repeated root $\lambda_0 = -\frac{b}{2a}$. Dividing the recurrence by a shows that it takes the form

$$u_n - 2\lambda_0 u_{n-1} + \lambda_0^2 u_{n-2} = 0.$$

Here $u_n = \lambda_0^n$ is one solution, but we need a second independent solution to accommodate two initial conditions. Let us test $u_n = n\lambda_0^n$ by substituting it directly into the left-hand side:

$$\begin{aligned} n\lambda_0^n - 2\lambda_0(n-1)\lambda_0^{n-1} + \lambda_0^2(n-2)\lambda_0^{n-2} &= \lambda_0^n[n - 2(n-1) + (n-2)] \\ &= \lambda_0^n(n - 2n + 2 + n - 2) = 0. \end{aligned}$$

Thus $n\lambda_0^n$ is indeed a second solution for all n. By the principle of superposition, the general solution for a repeated root λ_0 is

$$u_n = A\lambda_0^n + Bn\lambda_0^n = (A + Bn)\lambda_0^n.$$

Example 6

Question: Solve the recurrence relation $u_n - 6u_{n-1} + 9u_{n-2} = 0$ for $n \geq 2$, given the initial conditions $u_0 = 2$ and $u_1 = 12$.
Solution: Substituting $u_n = \lambda^n$ gives the auxiliary equation:

$$\lambda^2 - 6\lambda + 9 = 0 \implies (\lambda - 3)^2 = 0 \implies \lambda = 3 \text{ (repeated root)}.$$

Because $\lambda = 3$ is a repeated root, the general solution is

$$u_n = (A + Bn)3^n.$$

Substitute the initial conditions:

$$\begin{aligned} n = 0: &\quad (A+0)3^0 = 2 \implies A = 2 \\ n = 1: &\quad (A+B)3^1 = 12 \implies A + B = 4 \end{aligned}$$

Since $A = 2$, we have $2 + B = 4 \implies B = 2$. Therefore, the particular solution is

$$u_n = (2 + 2n)3^n = 2(n + 1)3^n.$$

Check ($n = 2$): From the recurrence, $u_2 = 6(12) - 9(2) = 54$. From the formula, $u_2 = 2(3)(3^2) = 54$.

3.2.6 Shifted Indices and Initial Conditions Starting at $n = 1$

A second-order recurrence relation is sometimes written as $au_{n+2} + bu_{n+1} + cu_n = 0$, or its initial conditions may be specified at $n = 1$ and $n = 2$ rather than $n = 0$ and $n = 1$. Substituting $u_n = \lambda^n$ and dividing by λ^n produces the exact same auxiliary equation $a\lambda^2 + b\lambda + c = 0$, and we simply substitute $n = 1$ and $n = 2$ into the general solution to find A and B.

Example 7

Question: A sequence satisfies $u_{n+2} + 2u_{n+1} - 3u_n = 0$ for $n \geq 1$, with $u_1 = 5$ and $u_2 = -7$. Find an explicit formula for u_n.
Solution: Substituting $u_n = \lambda^n$ gives $\lambda^{n+2} + 2\lambda^{n+1} - 3\lambda^n = 0$. Dividing by λ^n gives the auxiliary equation:

$$\lambda^2 + 2\lambda - 3 = 0 \implies (\lambda - 1)(\lambda + 3) = 0 \implies \lambda_1 = 1, \quad \lambda_2 = -3.$$

The general solution is

$$u_n = A(1)^n + B(-3)^n = A + B(-3)^n.$$

Using the initial conditions at $n = 1$ and $n = 2$:

$$\begin{aligned} n = 1: &\quad A - 3B = 5 && \text{— (Equation 1)} \\ n = 2: &\quad A + 9B = -7 && \text{— (Equation 2)} \end{aligned}$$

Subtracting Equation 1 from Equation 2 gives $12B = -12 \implies B = -1$. Substituting $B = -1$ into Equation 1 gives $A + 3 = 5 \implies A = 2$. Hence

$$u_n = 2 - (-3)^n.$$

Exercise 3B

1 For each of the following homogeneous recurrence relations, write down the auxiliary equation and find the general solution for u_n:

(a) $u_n - 8u_{n-1} + 15u_{n-2} = 0$ **[2 marks]**

(b) $u_n + 2u_{n-1} - 8u_{n-2} = 0$ **[2 marks]**

(c) $u_{n+2} - 10u_{n+1} + 25u_n = 0$ **[2 marks]**

2 Find the particular solution to the recurrence relation $u_n = 4u_{n-1} + 5u_{n-2}$ for $n \geq 2$, subject to the initial conditions $u_0 = 3$ and $u_1 = 3$. **[4 marks]**

3 Determine the closed-form formula for u_n given that $u_n - 4u_{n-1} + 4u_{n-2} = 0$ for $n \geq 2$, with $u_0 = 1$ and $u_1 = 6$. **[4 marks]**

4 A sequence is defined for $n \geq 1$ by $2u_{n+2} - 5u_{n+1} + 2u_n = 0$, with $u_1 = 5$ and $u_2 = 8.5$. Find an explicit formula for u_n in terms of n. **[5 marks]**

5 The Lucas numbers L_n satisfy the recurrence relation $L_n = L_{n-1} + L_{n-2}$ for $n \geq 2$, with initial conditions $L_0 = 2$ and $L_1 = 1$. Show that the closed-form solution for the Lucas numbers is given by

$$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n.$$

[5 marks]

6 ⋆ (a) A second-order homogeneous linear recurrence relation $u_n + bu_{n-1} + cu_{n-2} = 0$ has particular solution $u_n = (3-n)(-2)^n$ for $n \geq 0$. Find the values of the constants b and c, and state the values of u_0 and u_1. **[4 marks]**

(b) A student attempts to solve $v_n - 6v_{n-1} + 9v_{n-2} = 0$ with $v_0 = 2$ and $v_1 = 9$ by writing the trial general solution as $v_n = A(3)^n + B(3)^n$. Explain why the student's simultaneous equations for A and B have no solution. **[2 marks]**

3.3 Boolean Algebra and Propositional Logic

3.3.1 Propositions and Truth Values

In the mid-nineteenth century, George Boole developed an algebraic framework for analysing logical statements. In propositional logic, the basic building blocks are **propositions**: declarative statements that are unambiguously either **True** (T) or **False** (F), but not both.

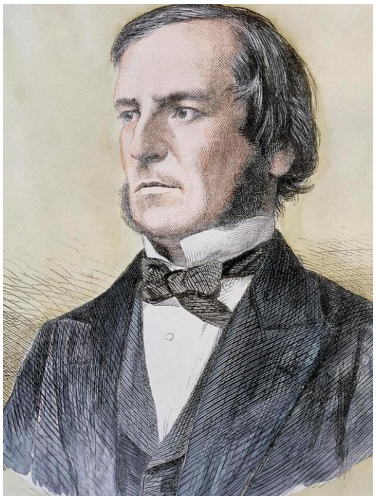


Figure 3.2: George Boole (1815–1864), the first Professor of Mathematics at Queen's College Cork, whose *Laws of Thought* (1854) set out the algebra of logic.

For example:

- "Belfast is the capital city of Northern Ireland." (A proposition with truth value T.)
- "The integer 7 is an even number." (A proposition with truth value F.)
- "The interior angles of any planar triangle sum to 180°." (A proposition with truth value T.)

By contrast, the following sentences are *not* propositions:

- "What time is the next train?" (A question cannot be assigned a truth value.)

- "Close the door." (An instruction or command has no truth value.)
- "This sentence is false." (A self-referential paradox: if it were true it would be false, and if it were false it would be true, so no consistent truth value exists.)

We use lower-case letters such as p, q and r to represent propositional variables. Each variable can take one of two truth values, written as T and F (or 1 and 0 in digital logic).

3.3.2 Logical Connectives and Precedence

Simple propositions can be combined using **logical operators** (or connectives) to form **compound propositions** whose truth values depend only on the truth values of their component variables.

Key Point

Fundamental Logical Operators:

- **Negation (NOT), written $\neg p$:** Reverses the truth value of p. Thus $\neg p$ is T when p is F, and F when p is T.
- **Conjunction (AND), written $p \wedge q$:** Evaluates to T if and only if **both** p and q are T; if at least one of p or q is F, then $p \wedge q$ is F.
- **Disjunction (OR), written $p \vee q$:** Evaluates to T if **at least one** of p or q is T, including the case where both p and q are T. It evaluates to F only when both p and q are F.

p	q	$\neg p$	$p \wedge q$	$p \vee q$
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

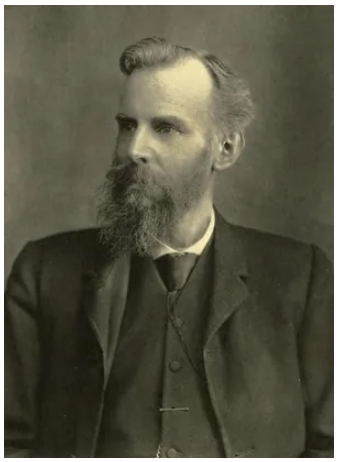


Figure 3.3: John Venn (1834–1923), who introduced his diagrams in 1880 to represent logical propositions.

These three operations correspond directly to set operations on Venn diagrams, where the region inside circle p represents the outcomes for which proposition p is true:

- Conjunction $p \wedge q$ corresponds to set intersection $P \cap Q$.

- Disjunction $p \vee q$ corresponds to set union $P \cup Q$.
- Negation $\neg p$ corresponds to the set complement P'.

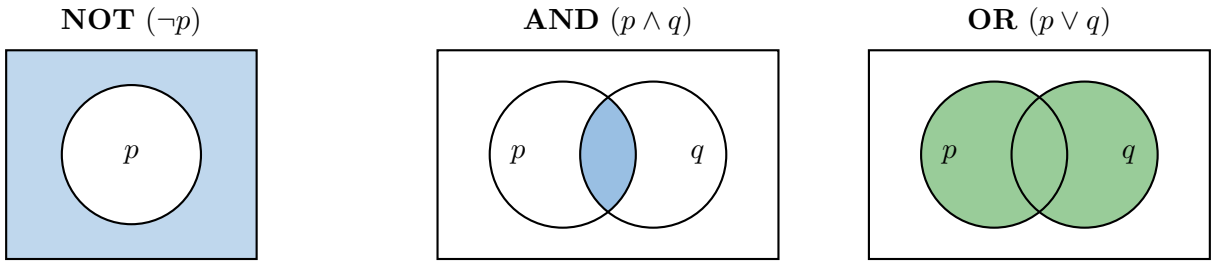


Figure 3.4: Venn diagram representations of $\neg p$, $p \wedge q$ and $p \vee q$. In each diagram, the shaded region indicates where the proposition evaluates to True.

To avoid ambiguity in compound expressions without brackets, logical operators follow a standard **order of precedence** (from highest to lowest):

1. Expressions inside brackets $(\dots)$
2. Negation $\neg$
3. Conjunction $\wedge$
4. Disjunction $\vee$

For instance, $\neg p \wedge q$ means $(\neg p) \wedge q$, where $\neg$ applies only to p, whereas $\neg(p \wedge q)$ negates the entire conjunction $p \wedge q$. Similarly, $\neg p \vee q \wedge r$ is evaluated as $(\neg p) \vee (q \wedge r)$.

3.3.3 Truth Tables and Proving Logical Equivalence

Two compound propositions P and Q are **logically equivalent**, written $P \equiv Q$, if they have the same truth value for every possible assignment of truth values to their underlying variables. Conversely, if there is even a single row of truth assignments for which P and Q take different truth values, then P and Q are **not** logically equivalent.

Because each variable can take 2 truth values (T or F), a truth table with n distinct variables has exactly 2^n rows:

- For 2 variables (p, q), there are $2^2 = 4$ rows.
- For 3 variables (p, q, r), there are $2^3 = 8$ rows.

To ensure that no combination is omitted or duplicated in an 8-row table, list the input columns in systematic binary counting order:

- Column p: four Ts followed by four Fs (T, T, T, T, F, F, F, F).
- Column q: alternating pairs of Ts and Fs (T, T, F, F, T, T, F, F).
- Column r: alternating single T and F (T, F, T, F, T, F, T, F).

Common Misconception

1. **Treating mathematical OR as exclusive:** In everyday speech, "tea or coffee" often means one or the other, not both. In mathematics, $p \vee q$ is always *inclusive*: when p is T and q is T, $p \vee q$ is T.
2. **Distributing $\neg$ without reversing the operator:** $\neg(p \vee q)$ is *not* equivalent to $\neg p \vee \neg q$, and $\neg(p \wedge q)$ is *not* equivalent to $\neg p \wedge \neg q$. Negating a bracketed conjunction or disjunction swaps $\wedge$ and $\vee$ (De Morgan's Laws).
3. **Omitting the concluding statement:** In an examination proof of equivalence, completing the truth table is not the final step. You must explicitly state that because the columns for the two propositions are identical in every row, the propositions are logically equivalent.

Example 8

Question: Construct a truth table to prove De Morgan's first law:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q.$$

Solution: With 2 variables (p and q), the truth table requires $2^2 = 4$ rows. We include intermediate columns for $p \vee q$, $\neg p$ and $\neg q$:

p	q	$p \vee q$	$\neg(\mathbf{p \vee q})$	$\neg p$	$\neg q$	$\neg \mathbf{p} \wedge \neg \mathbf{q}$
T	T	T	**F**	F	F	**F**
T	F	T	**F**	F	T	**F**
F	T	T	**F**	T	F	**F**
F	F	F	**T**	T	T	**T**

The truth values in the column for $\neg(p \vee q)$ and the column for $\neg p \wedge \neg q$ are identical (F, F, F, T) across all four rows. Therefore, $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Example 9

Question: Construct a truth table to prove the distributive law of conjunction over disjunction:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r).$$

Solution: Since there are 3 variables (p, q, r), we construct an 8-row truth table using the standard binary ordering:

p	q	r	$q \vee r$	$\mathbf{p \wedge (q \vee r)}$	$p \wedge q$	$p \wedge r$	$(\mathbf{p \wedge q}) \vee (\mathbf{p \wedge r})$
T	T	T	T	**T**	T	T	**T**
T	T	F	T	**T**	T	F	**T**
T	F	T	T	**T**	F	T	**T**
T	F	F	F	**F**	F	F	**F**
F	T	T	T	**F**	F	F	**F**
F	T	F	T	**F**	F	F	**F**
F	F	T	T	**F**	F	F	**F**
F	F	F	F	**F**	F	F	**F**

Because the columns for $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ are identical in all eight rows, we conclude that $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

Truth tables can also be used to show that two expressions are *not* logically equivalent, or to verify that a compound proposition simplifies to a shorter expression.

Example 10

Question:

(a) Explain, by referring to a specific assignment of truth values, why $\neg p \wedge q$ is **not** logically equivalent to $\neg(p \wedge q)$.

(b) Construct a truth table to prove that $\neg(p \wedge q) \wedge p \equiv p \wedge \neg q$.

Solution:

(a) When p is T and q is F, $\neg p \wedge q$ evaluates to $\mathrm{F} \wedge \mathrm{F} = \mathrm{F}$, whereas $\neg(p \wedge q)$ evaluates to $\neg(\mathrm{T} \wedge \mathrm{F}) = \neg(\mathrm{F}) = \mathrm{T}$. Because their truth values differ for this assignment, $\neg p \wedge q \not\equiv \neg(p \wedge q)$.

(b) Constructing the 4-row truth table for $\neg(p \wedge q) \wedge p$ and $p \wedge \neg q$:

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg(\mathbf{p \wedge q}) \wedge \mathbf{p}$	$\neg q$	$\mathbf{p \wedge \neg q}$
T	T	T	F	**F**	F	**F**
T	F	F	T	**T**	T	**T**
F	T	F	T	**F**	F	**F**
F	F	F	T	**F**	T	**F**

The columns for $\neg(p \wedge q) \wedge p$ and $p \wedge \neg q$ are identical (F, T, F, F) across all four rows, proving that $\neg(p \wedge q) \wedge p \equiv p \wedge \neg q$.

Beyond the Syllabus: Logic Gates, Hardware, and Real-World "Truthiness"

In modern computer hardware, boolean propositions are represented by electrical voltages (High voltage = True, Low voltage = False). Engineers use physical semiconductor devices called **Logic Gates** to perform boolean operations. An AND gate takes two input wires and produces a high output voltage only if both input wires are high. An OR gate produces a high output if at least one input wire is high. By wiring thousands, or even millions, of these simple gates together in intricate configurations, computer processors can evaluate immensely complex boolean expressions, allowing them to perform arithmetic, render graphics, and execute software instructions at lightning speed. The *software* built on top of this hardware is rarely so disciplined. In this chapter every proposition is strictly **True** or **False**, and the operators $\wedge$, $\vee$ and $\neg$ are only ever applied to those two values — nothing is silently converted from one kind of thing into another. Most programming languages, by contrast, introduce **"truthiness"**: non-Boolean values (numbers, strings, empty lists) are implicitly coerced to `true` or `false` whenever they appear in a condition.

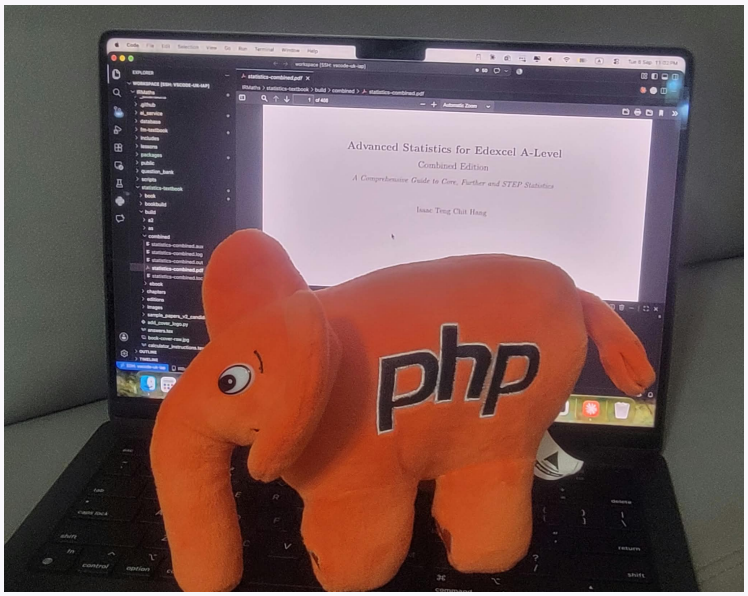


Figure 3.5: The elePHPant, PHP's mascot, posing for a photo with a well-thumbed statistics textbook — a cheerful reminder to mind your `===`.

The language PHP is infamous for this. Its **loose** comparison operator `==`, which coerces types *before* comparing, produces results that would horrify a logician:

- `"0" == false` evaluates to `true`;
- `null == false` evaluates to `true`;
- `0 == ""` and `"abc" == 0` were even `true` in older versions of the language.

PHP's loose-comparison relation is not always transitive: it is possible to have `a == b` and `b == c` while `a == c` is false. That cannot happen with mathematical equality.
The safer alternative is the **strict** equality operator `===`, which compares both *value and type* without coercion. This is not the same relation as logical equivalence $\equiv$, but it reflects the same discipline: the meaning and type of the objects being compared must be fixed before a valid comparison can be made.

Exercise 3C

1 Copy and complete the following truth table, and hence prove the Absorption Law $p \vee (p \wedge q) \equiv p$.

p	q	$p \wedge q$	$p \vee (p \wedge q)$
T	T		
T	F		
F	T		
F	F		

[3 marks]

2 Construct a complete truth table to prove De Morgan's second law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$. **[4 marks]**

3 Use a truth table to prove that $(p \vee q) \wedge (\neg p \vee q) \equiv q$. **[4 marks]**

4 Use a systematic truth table with three variables to prove the distributive law of disjunction over conjunction:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r).$$

[6 marks]

5 The exclusive OR operator, written $p \oplus q$, is defined to be True when **exactly one** of p or q is True, and False when both are True or both are False.

(a) Construct a truth table to prove that $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$. **[4 marks]**

(b) State, with a reason from your truth table, whether $p \oplus q$ is logically equivalent to $\neg(p \wedge q)$. **[1 mark]**

6 ⋆ (a) A student claims that $p \vee (q \wedge \neg r)$ is logically equivalent to $(p \vee q) \wedge \neg r$. By finding a combination of truth values for p, q and r for which the two propositions evaluate to different truth values, show that the student's claim is false. **[2 marks]**

(b) Construct a truth table with three variables to prove that $\neg(p \vee q \vee r) \equiv \neg p \wedge \neg q \wedge \neg r$. **[5 marks]**

End-of-Chapter Mixed Exercises

1. A homogeneous linear recurrence relation is given by $u_n - 9u_{n-1} + 20u_{n-2} = 0$ for $n \geq 2$. Find the general solution for u_n. **[3 marks]**

2. Solve the recurrence relation $u_n + u_{n-1} - 12u_{n-2} = 0$ for $n \geq 2$, subject to the initial conditions $u_0 = 5$ and $u_1 = 1$. **[5 marks]**

3. Find the particular solution to the recurrence relation $u_n - 10u_{n-1} + 25u_{n-2} = 0$ for $n \geq 2$, given that $u_0 = 3$ and $u_1 = 20$. **[5 marks]**

4. A sequence is generated by $u_n = 4u_{n-2}$ for $n \geq 2$. Given that $u_0 = 8$ and $u_1 = 4$, find the explicit formula for u_n and hence determine the value of u_{10}. **[4 marks]**

5. Construct a truth table to determine whether the implication proposition $p \Rightarrow q$ is logically equivalent to $\neg p \vee q$. (Note: $p \Rightarrow q$ is False when p is True and q is False, and True in all other cases.) **[3 marks]**

6. Using a truth table, prove that $\neg(p \vee q) \vee (\neg p \wedge q) \equiv \neg p$. **[5 marks]**

7. Construct a truth table to prove the associative law of conjunction: $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$. **[5 marks]**

8. The sequence u_n is defined by $u_n = pu_{n-1} + qu_{n-2}$ for $n \geq 2$, where p and q are constants. Given that the general solution is $u_n = A(2)^n + B(-3)^n$, find the values of p and q. **[4 marks]**

9. A sequence is defined by $u_{n+2} = 2u_{n+1} - u_n$ for $n \geq 1$. Given that $u_1 = 4$ and $u_2 = 7$, find an explicit simplified formula for u_n in terms of n. **[5 marks]**

10. A discrete population model predicts that the population P_n in year n satisfies $P_n = 1.6P_{n-1} - 0.64P_{n-2}$ for $n \geq 2$. Given that $P_0 = 100$ and $P_1 = 120$, find an explicit expression for P_n in terms of n and calculate the predicted population in year 10 to the nearest whole number. **[6 marks]**

11. $\star$ A commercial fish farm stocks a tank with 2000 trout at the start of year 0. During each year, the number of trout increases by 20% through breeding, and at the end of the year 350 trout are harvested for sale. Let F_n denote the number of trout in the tank at the end of year n.

    (a) Write down a recurrence relation for F_n and state the initial condition. **[2 marks]**

    (b) Calculate the number of trout in the tank at the end of year 3. **[2 marks]**

    (c) If the farm instead harvested H trout at the end of each year, find the smallest integer value of H for which the trout population would decrease from year to year. **[2 marks]**

12. A Fibonacci-type sequence satisfies the recurrence relation $u_n = 2u_{n-1} + u_{n-2}$ for $n \geq 2$, with initial conditions $u_0 = 0$ and $u_1 = 2$. Find the exact closed-form solution for u_n in terms of n. **[5 marks]**

13. A machine has 1000 serviceable components initially. Each month 10% of the current serviceable components fail, then 80 replacements are installed. Let $u_0 = 1000$.

    (a) Form a recurrence, find the equilibrium and compute u_1, u_2. **[4 marks]**

    (b) Put $v_n = u_n - 800$. Find u_n explicitly and explain a limitation of the model. **[6 marks]**

14. A sequence satisfies $u_{n+2} = 4u_{n+1} - 4u_n$, $u_0 = 3$, $u_1 = 8$.

    (a) Find an explicit solution and verify it at $n = 2$. **[6 marks]**

(b) A student writes $u_n = 3 \cdot 2^n + 2n$. Check both initial conditions and explain why this does not establish a solution. **[4 marks]**

15. Compare the safety rules $A = (p \lor q) \land \neg r$ and $B = p \lor (q \land \neg r)$.

(a) Construct a truth table for both rules. **[6 marks]**

(b) If p, q are alternative permissions and r is an emergency stop, decide which rule enforces the stop and give a counterexample for the other. **[4 marks]**

Chapter 3 Summary

Summary of Key Formulas and Concepts

1. **Modelling with Recurrence Relations:**
   - A recurrence relation defines u_n in terms of preceding terms (u_{n-1} for first-order; u_{n-1} and u_{n-2} for second-order).
   - Initial conditions must be provided (1 value for first-order, 2 values for second-order) to specify a unique sequence.
   - In applied models, check whether u_0 or u_1 represents the starting state and verify the order in which percentage changes and fixed additions/removals occur.
2. **Solving Homogeneous Linear Recurrence Relations with Constant Coefficients:**
   - Standard second-order form: $au_n + bu_{n-1} + cu_{n-2} = 0$ (or $au_{n+2} + bu_{n+1} + cu_n = 0$).
   - Substituting $u_n = \lambda^n$ ($\lambda \neq 0$) gives the quadratic **auxiliary equation** $a\lambda^2 + b\lambda + c = 0$.
   - **Distinct real roots** ($\lambda_1 \neq \lambda_2$): $u_n = A\lambda_1^n + B\lambda_2^n$.
   - **Repeated real root** ($\lambda_1 = \lambda_2 = \lambda$): $u_n = (A + Bn)\lambda^n$.
   - **Fibonacci-type recurrences:** When the auxiliary equation has irrational roots (such as $\lambda = \frac{1\pm\sqrt{5}}{2}$ for $u_n = u_{n-1} + u_{n-2}$), retain exact surds when solving for A and B.
3. **Boolean Algebra and Propositional Logic:**
   - A **proposition** is a declarative statement that is unambiguously True (T) or False (F).
   - **Operators:** Negation $\neg p$ (True when p is False); Conjunction $p \wedge q$ (True only when both p and q are True); Inclusive Disjunction $p \vee q$ (True when at least one of p or q is True).
   - **Order of precedence:** Brackets first, then $\neg$, then $\wedge$, then $\vee$.
   - **Logical equivalence ($P \equiv Q$):** Proven by constructing a complete truth table ($2^2 = 4$ rows for 2 variables; $2^3 = 8$ rows for 3 variables in binary order 4T/4F, 2T/2F, 1T/1F) and stating that the columns for P and Q are identical in every row.

Appendix A

Calculator Guide

This appendix gives key sequences for three calculator models:

- **Casio ClassWiz fx-991EX**;
- **Casio ClassWiz fx-991CW**;
- **Casio fx-CG50** graphical calculator.

Menu wording can differ slightly after an operating-system update. Practise these procedures on your own calculator before the examination.

Key Point

Calculator output supports a written solution. Show the recurrence and its initial conditions, algorithm steps, simplex row operations or cycle-index substitution before quoting a result. Keep exact values until the final numerical answer wherever possible.

A.1 Counting and Restricted Assignments

For an unordered selection use $\binom{n}{r}$, not ${}^{n}P_{r}$. In a rook-polynomial calculation, find the coefficients on paper before evaluating the alternating factorial sum.

Casio fx-991EX

1. Press **MENU**, select **Calculate**, then press =.
2. For $\binom{10}{3}$, enter **10**, **SHIFT**, the division key (**nCr**), **3**, then =. The result is 120.
3. For 5!, enter **5**, **SHIFT**, the reciprocal key ($x!$), then =. The result is 120.

Casio fx-991CW

1. Press **HOME**, choose **Calculate**, then **OK**.
2. For $\binom{10}{3}$, enter **10**, then **CATALOG**, **Probability**, **Combination(C)**; enter **3** and press **EXE**.
3. For 5!, enter **5**, then **CATALOG**, **Probability**, **Factorial(!)**, and **EXE**.

Casio fx-CG50

1. From **MENU**, open **Run-Matrix**.
2. Enter **10**, press **OPTN**, then the next-page soft key **F6** (▷), and **PROB**. Select **nCr**, enter **3**, then **EXE**.
3. For 5!, enter **5**, open the same **PROB** menu, choose $x!$, then **EXE**.

For a forbidden-board polynomial $1 + 4x + 4x^2 + x^3$ on a 4×4 board, evaluate $4! - 4(3!) + 4(2!) - 1! = 7$. A count must be a non-negative integer; this is a useful check, not a proof that the rook coefficients are correct.

A.2 Recurrence Relations

Casio fx-991EX: first-order iteration

1. In **Calculate**, enter the initial value **500**, then =.
2. Enter `0.8×Ans+60`, using the **Ans** key, and press =.
3. Repeat = to apply the same expression to the latest answer. Record the index and value after each step: 460, 428, 402.4.

Casio fx-991CW: first-order iteration

1. In **Calculate**, enter **500**, then **EXE**.
2. Enter `0.8×Ans+60` and press **EXE**.
3. Press **EXE** again to repeat the expression with the updated answer. Record the term number alongside each result.

Casio fx-CG50: recurrence table

1. From **MENU**, open **Recursion**. Press **F3** (TYPE), then **F2** (a_{n+1}).
2. Enter $0.8a_n + 60$ in the a_{n+1} line, using the on-screen a_n soft key; press **EXE**.
3. Press **F5** (SET). Choose a table starting at 0, ending at 3, with $a_0 = 500$. Press **EXE** after each entry.
4. Return with **EXIT**, then press **F6** (TABLE). The entries should be $500, 460, 428, 402.4$ at $n = 0, 1, 2, 3$.

Key Point

A second-order recurrence needs both previous values. A single **Ans** register does not retain that pair. Record successive terms explicitly or use an appropriate recurrence table. Check a closed form at the first term not used to find its constants.

A.3 PERT Means, Variances and Probabilities

Calculate each activity's expected duration and variance before identifying the critical path:

$$\mu = \frac{a + 4m + b}{6}, \qquad \sigma^2 = \left(\frac{b - a}{6}\right)^2.$$

For independent activities on the selected path, add means and variances. If the normal approximation is $T \approx N(11, 2)$, the standard deviation is $\sqrt{2}$, not 2.

Casio fx-991EX

1. Press **MENU**, **7: Distribution**, then choose **Normal CD**.
2. Enter Lower, Upper, σ, μ. For $P(T \leq 13)$ use a sufficiently low lower bound such as -10^{99}, Upper 13, $\sigma = \sqrt{2}$, $\mu = 11$.
3. Press $=$ to obtain approximately 0.921. For a deadline from a probability, choose **Inverse Normal** and enter the area to the left, σ and μ.

Casio fx-991CW

1. Press **HOME**, select **Distribution**, then **Normal CD** and **Variable**.
2. Enter Lower -10^{99}, Upper 13, $\sigma = \sqrt{2}$, $\mu = 11$; press **EXE**.
3. For a deadline from a probability, select **Inverse Normal**; enter the left-tail area and the path's standard deviation and mean.

Casio fx-CG50

1. From **Statistics**, press **F5** (DIST), **F1** (NORM), **F2** (Ncd).
2. Enter Lower -10^{99}, Upper 13, $\sigma = \sqrt{2}$, $\mu = 11$; press **EXE**.
3. For an inverse calculation, use **F3** (InvN), select the left tail and enter the area, standard deviation and mean.

The fixed-path normal calculation is an approximation. A competing path can become critical in an actual project; calculator output does not remove this modelling limitation.

A.4 Simplex and Cycle-Index Arithmetic

ClassWiz: exact row calculations

1. Use **Calculate** mode. Divide every entry of the pivot row, including its RHS, by the pivot. Retain fractions rather than rounded decimals.
2. To eliminate a coefficient c, calculate each entry of $R - cR_p$, where R_p is the normalised pivot row. Write the resulting complete tableau.
3. Check the pivot column is a unit column, then substitute the final variables into the original constraints and profit expression.

Casio fx-CG50: exact row calculations

1. Use **Run-Matrix** for the same arithmetic. Keep a fixed column order and perform the written pivot operations on every entry, including the objective row.
2. Check the final basic variables against the original constraints. Whole-unit requirements must be checked separately from the continuous optimum.

For cycle indices, calculate each class contribution separately before summing and dividing by the group order. Under square rotations, two red and two blue vertices give $(6+2+0)/4 = 2$. The coefficients must be found from the pattern inventory on paper; entering 2 for each x_i would instead count every composition from a two-colour palette.

Final Calculator Checklist

Before the examination, practise:

- distinguishing unordered selections from permutations;
- entering brackets around negative values and fractions;
- recording both previous terms for a second-order recurrence;
- using standard deviation, rather than variance, in a Normal-CD command;
- selecting the correct normal tail or inverse-normal area;
- keeping exact fractions through simplex pivots;
- checking algorithm steps and the meaning of the final numerical answer.

AS Discrete and Decision Mathematics Practice Set 1

CCEA Further Mathematics Discrete and Decision Mathematics

AS Discrete and Decision Mathematics Practice Set 1

Level: AS
Total marks: 50

Instructions

- Answer all questions in this Discrete and Decision Mathematics section.
- A calculator may be used. Show sufficient working to make your method clear.
- Give numerical answers to three significant figures unless otherwise stated.
- The marks for each question or part are shown in brackets.

1. Let $G = \{1, 2, 4, 5, 7, 8\}$ under multiplication modulo 9.
   (a) Given that G is a group, find the successive powers and period of 2. **[3 marks]**
   (b) Find all generators of G and the subgroup generated by 4. **[3 marks]**
   (c) A student claims that G is isomorphic to the symmetry group of an equilateral triangle because both have order 6. Explain the error. **[4 marks]**
2. A connected undirected network has edges $AB = 4$, $AC = 7$, $BC = 2$, $BD = 5$, $CD = 1$, $CE = 6$, $DE = 3$.
   (a) Apply Prim's algorithm from A, recording the edges in order and the total weight. **[5 marks]**
   (b) Apply Dijkstra's algorithm from A. Record permanent labels in order, and find a shortest route to E. **[5 marks]**
3. A project has activities A, B, C, D, E with durations $3, 5, 4, 2, 3$ days respectively. A and B may start immediately; C follows A; D follows both A and B; E follows both C and D.
   (a) Draw an activity network satisfying the precedence table. **[3 marks]**
   (b) Find earliest and latest start times for all activities, the project duration and every critical path. **[5 marks]**
   (c) Explain the effect of delaying activity B by one day. **[2 marks]**
4. A sequence satisfies $u_{n+2} = 5u_{n+1} - 6u_n$, with $u_0 = 2$ and $u_1 = 5$.
   (a) Find an explicit expression for u_n. **[6 marks]**
   (b) Calculate u_4 both from the recurrence and from your expression. **[4 marks]**
5. An alarm is active when $(p \wedge q) \vee (p \wedge \neg q)$ is true.
   (a) Construct a truth table and identify an equivalent proposition involving only one variable. **[6 marks]**
   (b) Explain whether changing q alone can change the alarm state, and why $p \wedge (q \vee \neg q)$ agrees with your answer. **[4 marks]**

END OF SECTION
TOTAL FOR SECTION IS 50 MARKS

Answers

Recurrences and Boolean Algebra

Exercise 3A

1. (a) Substituting successively into $u_n = 3u_{n-1} - 2$ from $u_1 = 4$:

$$u_2 = 3(4) - 2 = 10$$
$$u_3 = 3(10) - 2 = 28$$
$$u_4 = 3(28) - 2 = 82$$

   (b) Working backwards from $v_3 = 19$:

$$3v_2 - 2 = 19 \implies 3v_2 = 21 \implies v_2 = 7$$

$$3v_1 - 2 = 7 \implies 3v_1 = 9 \implies v_1 = 3$$

2. (a) The initial condition is $A_0 = 12$. After 2 weeks, $A_2 = kA_1 = k^2A_0 = 12k^2$. Setting $12k^2 = 27$ gives $k^2 = \frac{27}{12} = \frac{9}{4}$. Since $k > 0$, $k = \frac{3}{2} = 1.5$.

   (b) At the end of week 5:

$$A_5 = A_2 \times k^3 = 27 \times (1.5)^3 = 27 \times \frac{27}{8}$$
$$= \frac{729}{8} \text{ m}^2 = 91.125 \text{ m}^2$$
$$\text{(or } 91.1 \text{ m}^2 \text{ to 3 s.f.)}$$

3. (a) Since 40% of the drug is removed over 24 hours, 60% (0.6) remains in the bloodstream before the next 50 mg tablet is taken:

$$D_n = 0.6D_{n-1} + 50 \quad (n \geq 2), \quad D_1 = 50$$

   (b) Iterating from $D_1 = 50$:

$$D_2 = 0.6(50) + 50 = 80$$
$$D_3 = 0.6(80) + 50 = 98$$
$$D_4 = 0.6(98) + 50 = 108.8 \text{ mg}$$

   (c) Solving $L = 0.6L + 50$ gives $0.4L = 50 \implies L = 125$ mg. Since D_n increases towards the upper limit of 125 mg and $125 < 130$, the mass of the drug in the bloodstream never exceeds 130 mg.

4. (a) Under Model A, the 5% annual growth increases the population by a factor of 1.05 before h trees are felled:

$$T_n = 1.05T_{n-1} - h \quad (n \geq 1),$$
$$T_0 = 4000$$

   (b) Under Model B, felling h trees at the start of year n leaves $T_{n-1} - h$ trees. Applying the 5% growth to the remaining trees gives

$$T_n = 1.05(T_{n-1} - h) = 1.05T_{n-1} - 1.05h$$

(c) Using Model A with $T_0 = 4000$:

$$T_1 = 1.05(4000) - h = 4200 - h$$
$$T_2 = 1.05(4200 - h) - h = 4410 - 2.05h$$

Setting $T_2 = 3959$ gives $4410 - 2.05h = 3959 \implies 2.05h = 451 \implies h = 220$ trees.

5. (a) For 1 step there is 1 move [1], so $s_1 = 1$. For 2 steps the possible sequences are $[1, 1]$ and [2], so $s_2 = 2$.

(b) For $n \geq 3$ steps, the drone's first move is either 1 step (leaving $n - 1$ steps, which can be climbed in s_{n-1} ways) or 2 steps (leaving $n - 2$ steps, which can be climbed in s_{n-2} ways). Because these two cases are mutually exclusive and cover all possibilities, $s_n = s_{n-1} + s_{n-2}$ for $n \geq 3$. Generating the terms up to $n = 7$:

$$s_3 = 2 + 1 = 3, \quad s_4 = 3 + 2 = 5,$$
$$s_5 = 5 + 3 = 8, \quad s_6 = 8 + 5 = 13,$$
$$s_7 = 13 + 8 = 21$$

Exercise 3B

1. (a) Auxiliary equation: $\lambda^2 - 8\lambda + 15 = 0 \implies (\lambda - 3)(\lambda - 5) = 0 \implies \lambda = 3, 5$.
General solution: $u_n = A(3)^n + B(5)^n$.

(b) Auxiliary equation: $\lambda^2 + 2\lambda - 8 = 0 \implies (\lambda + 4)(\lambda - 2) = 0 \implies \lambda = 2, -4$.
General solution: $u_n = A(2)^n + B(-4)^n$.

(c) Auxiliary equation: $\lambda^2 - 10\lambda + 25 = 0 \implies (\lambda - 5)^2 = 0 \implies \lambda = 5$ (repeated root).
General solution: $u_n = (A + Bn)5^n$.

2. Rearranging gives $u_n - 4u_{n-1} - 5u_{n-2} = 0$. The auxiliary equation is

$$\lambda^2 - 4\lambda - 5 = 0 \implies (\lambda - 5)(\lambda + 1) = 0$$
$$\implies \lambda_1 = 5, \quad \lambda_2 = -1$$

The general solution is $u_n = A(5)^n + B(-1)^n$. Substituting $u_0 = 3$ and $u_1 = 3$:

$$n = 0: \quad A + B = 3$$
$$n = 1: \quad 5A - B = 3$$

Adding the two equations gives $6A = 6 \implies A = 1$, and hence $B = 3 - 1 = 2$. The particular solution is

$$u_n = 5^n + 2(-1)^n$$

3. The auxiliary equation is

$$\lambda^2 - 4\lambda + 4 = 0 \implies (\lambda - 2)^2 = 0$$
$$\implies \lambda = 2 \text{ (repeated root)}$$

The general solution is $u_n = (A + Bn)2^n$. Substituting $u_0 = 1$ and $u_1 = 6$:

$$n = 0: \quad (A + 0)2^0 = 1 \implies A = 1$$
$$n = 1: \quad (A + B)2^1 = 6$$
$$\implies A + B = 3 \implies B = 2$$

The closed-form formula is $u_n = (1 + 2n)2^n$.

4. Substituting $u_n = \lambda^n$ gives the auxiliary equation

$$\begin{aligned} 2\lambda^2 - 5\lambda + 2 = 0 &\implies (2\lambda - 1)(\lambda - 2) = 0 \\ &\implies \lambda_1 = 2, \quad \lambda_2 = \frac{1}{2} \end{aligned}$$

The general solution is $u_n = A(2)^n + B\left(\frac{1}{2}\right)^n$. Using $u_1 = 5$ and $u_2 = 8.5 = \frac{17}{2}$:

$$\begin{aligned} n = 1: &\quad 2A + \frac{1}{2}B = 5 \implies 4A + B = 10 \\ n = 2: &\quad 4A + \frac{1}{4}B = \frac{17}{2} \implies 16A + B = 34 \end{aligned}$$

Subtracting the first equation from the second gives $12A = 24 \implies A = 2$, and substituting back gives $8 + B = 10 \implies B = 2$. Hence

$$u_n = 2(2)^n + 2\left(\frac{1}{2}\right)^n = 2^{n+1} + 2^{1-n}$$

5. Rearranging gives $L_n - L_{n-1} - L_{n-2} = 0$, with auxiliary equation $\lambda^2 - \lambda - 1 = 0$. By the quadratic formula,

$$\lambda = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

Thus the general solution is

$$L_n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n$$

Applying $L_0 = 2$ gives $A + B = 2$. Applying $L_1 = 1$ gives

$$\begin{aligned} A\left(\frac{1+\sqrt{5}}{2}\right) + B\left(\frac{1-\sqrt{5}}{2}\right) &= 1 \\ \implies \frac{A+B}{2} + \frac{(A-B)\sqrt{5}}{2} &= 1 \end{aligned}$$

Substituting $A + B = 2$ yields $1 + \frac{(A-B)\sqrt{5}}{2} = 1 \implies (A-B)\sqrt{5} = 0 \implies A = B$. Since $A + B = 2$ and $A = B$, we obtain $A = 1$ and $B = 1$, giving

$$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$$

6. (a) Since $u_n = (3-n)(-2)^n$ has the repeated-root form $(A + Bn)\lambda^n$, the auxiliary equation has a repeated root at $\lambda = -2$: $(\lambda + 2)^2 = 0 \implies \lambda^2 + 4\lambda + 4 = 0$. Comparing with $\lambda^2 + b\lambda + c = 0$ gives $b = 4$ and $c = 4$. Substituting $n = 0$ and $n = 1$ gives $u_0 = (3-0)(-2)^0 = 3$ and $u_1 = (3-1)(-2)^1 = -4$.

   (b) The student's trial form $v_n = A(3)^n + B(3)^n = (A + B)3^n$ collapses into a single unknown constant $A + B$ because $\lambda = 3$ is a repeated root. Substituting $v_0 = 2$ gives $A + B = 2$, while substituting $v_1 = 9$ gives $3(A + B) = 9 \implies A + B = 3$; these two equations are inconsistent, whereas the correct general solution is $v_n = (A + Bn)3^n$.

Exercise 3C

1. Completing the truth table for $p \vee (p \wedge q)$:

$\mathbf{p}$	q	$p \wedge q$	$\mathbf{p \vee (p \wedge q)}$
T	T	T	**T**
T	F	F	**T**
F	T	F	**F**
F	F	F	**F**

   Since the column for $p \vee (p \wedge q)$ and the column for p are identical (T, T, F, F) across all four rows, $p \vee (p \wedge q) \equiv p$.

2. Constructing the 4-row truth table for $\neg(p \wedge q)$ and $\neg p \vee \neg q$:

p	q	$p \wedge q$	$\mathbf{\neg(p \wedge q)}$	$\neg p$	$\neg q$	$\mathbf{\neg p \vee \neg q}$
T	T	T	**F**	F	F	**F**
T	F	F	**T**	F	T	**T**
F	T	F	**T**	T	F	**T**
F	F	F	**T**	T	T	**T**

   Because the columns for $\neg(p \wedge q)$ and $\neg p \vee \neg q$ are identical (F, T, T, T) in all four rows, $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

3. Constructing the 4-row truth table for $(p \vee q) \wedge (\neg p \vee q)$ and q:

p	$\mathbf{q}$	$\neg p$	$p \vee q$	$\neg p \vee q$	$\mathbf{(p \vee q) \wedge (\neg p \vee q)}$
T	**T**	F	T	T	**T**
T	**F**	F	T	F	**F**
F	**T**	T	T	T	**T**
F	**F**	T	F	T	**F**

   Since the columns for $(p \vee q) \wedge (\neg p \vee q)$ and q are identical (T, F, T, F) in every row, $(p \vee q) \wedge (\neg p \vee q) \equiv q$.

4. Constructing the 8-row truth table for $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$:

p	q	r	$q \wedge r$	$\mathbf{p \vee (q \wedge r)}$	$p \vee q$	$p \vee r$	$\mathbf{(p \vee q) \wedge (p \vee r)}$
T	T	T	T	**T**	T	T	**T**
T	T	F	F	**T**	T	T	**T**
T	F	T	F	**T**	T	T	**T**
T	F	F	F	**T**	T	T	**T**
F	T	T	T	**T**	T	T	**T**
F	T	F	F	**F**	T	F	**F**
F	F	T	F	**F**	F	T	**F**
F	F	F	F	**F**	F	F	**F**

   Because the columns for $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are identical in all eight rows, $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

5. (a) Constructing the 4-row truth table for $p \oplus q$ and $(p \vee q) \wedge \neg(p \wedge q)$:

p	q	$\mathbf{p \oplus q}$	$p \vee q$	$p \wedge q$	$\neg(p \wedge q)$	$\mathbf{(p \vee q) \wedge \neg(p \wedge q)}$
T	T	**F**	T	T	F	**F**
T	F	**T**	T	F	T	**T**
F	T	**T**	T	F	T	**T**
F	F	**F**	F	F	T	**F**

Since the columns for $p \oplus q$ and $(p \vee q) \wedge \neg(p \wedge q)$ are identical (F, T, T, F) in all four rows, $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$.

(b) $p \oplus q$ is not logically equivalent to $\neg(p \wedge q)$ because when p is F and q is F (row 4), $p \oplus q$ is F whereas $\neg(p \wedge q)$ is T.

6. (a) Take $p = \mathrm{T}$, $q = \mathrm{T}$ and $r = \mathrm{T}$ (or $p = \mathrm{T}, q = \mathrm{F}, r = \mathrm{T}$). Then $\neg r = \mathrm{F}$, so

$$p \vee (q \wedge \neg r) = \mathrm{T} \vee (\mathrm{T} \wedge \mathrm{F}) = \mathrm{T} \vee \mathrm{F} = \mathrm{T}$$

whereas

$$(p \vee q) \wedge \neg r = (\mathrm{T} \vee \mathrm{T}) \wedge \mathrm{F} = \mathrm{T} \wedge \mathrm{F} = \mathrm{F}$$

Since the two propositions take different truth values for this assignment, the student's claim is false.

(b) Constructing the 8-row truth table for $\neg(p \vee q \vee r)$ and $\neg p \wedge \neg q \wedge \neg r$:

p	q	r	$p \vee q \vee r$	$\mathbf{\neg(p \vee q \vee r)}$	$\neg p$	$\neg q$	$\neg r$	$\mathbf{\neg p \wedge \neg q \wedge \neg r}$
T	T	T	T	**F**	F	F	F	**F**
T	T	F	T	**F**	F	F	T	**F**
T	F	T	T	**F**	F	T	F	**F**
T	F	F	T	**F**	F	T	T	**F**
F	T	T	T	**F**	T	F	F	**F**
F	T	F	T	**F**	T	F	T	**F**
F	F	T	T	**F**	T	T	F	**F**
F	F	F	F	**T**	T	T	T	**T**

Because the columns for $\neg(p \vee q \vee r)$ and $\neg p \wedge \neg q \wedge \neg r$ are identical (F, F, F, F, F, F, F, T) across all eight rows, $\neg(p \vee q \vee r) \equiv \neg p \wedge \neg q \wedge \neg r$.

AS Discrete and Decision Mathematics Practice Set 1 Solutions

1. (a) $2, 4, 8, 7, 5, 1$ modulo 9, so the period is 6. [M1 A1 A1]

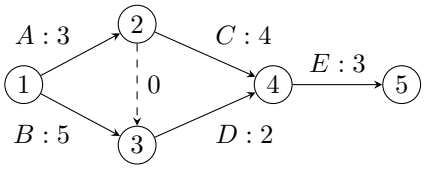
(b) The generators are 2 and $5 = 2^5$. The subgroup is $\langle 4 \rangle = \{1, 4, 7\}$. [B1 B1 A1]

(c) Equal order is necessary but not sufficient. Here G is cyclic and abelian; triangle symmetries are non-abelian and have no element of period 6. Either invariant disproves the claimed isomorphism. [B1 B1 B1 B1]

2. (a) Add AB, BC, CD, DE, with weights $4, 2, 1, 3$; total 10. Each edge joins the current tree to an unvisited vertex with the smallest available weight. [M1 A1 A1 A1 A1]

(b) Permanent labels: $A : 0$, $B : 4$, $C : 6$, $D : 7$, $E : 10$. A shortest route is $A - B - C - D - E$, of length 10. [M1 A1 A1 A1 A1]

3. (a) One activity-on-arc network is shown below. The dashed zero-duration dummy ensures that D follows both A and B, while C follows only A.



[B1 B1 B1]

(b) Earliest starts for A, B, C, D, E are $0, 0, 3, 5, 7$. Backwards from finish 10, latest starts are also $0, 0, 3, 5, 7$. Duration 10 days; both $A - C - E$ and $B - D - E$ are critical. [M1 M1 A1 A1 A1]

(c) $B - D - E$ then takes 11 days. Since B has zero float, completion is delayed by one day. [B1 B1]

4. (a) The auxiliary equation is $r^2 - 5r + 6 = (r - 2)(r - 3) = 0$. Thus $u_n = A2^n + B3^n$. From $A + B = 2$ and $2A + 3B = 5$, $A = B = 1$, so $u_n = 2^n + 3^n$. [M1 M1 M1 A1 A1 A1]

(b) The recurrence gives $u_2 = 13$, $u_3 = 35$, $u_4 = 97$. The formula gives $2^4 + 3^4 = 16 + 81 = 97$. [M1 A1 M1 A1]

5. (a) For $(p, q) = (T, T), (T, F), (F, T), (F, F)$, the two conjunction columns are (T, F, F, F) and (F, T, F, F). Their disjunction is (T, T, F, F), exactly the column for p. [M1 M1 M1 A1 A1 A1]

(b) Changing q cannot change the output, which is p. Since $q \vee \neg q$ is always true, the factored expression also reduces to p. [B1 B1 B1 B1]